1421 Research Park Dr
Lawrence, KS 66049

Long Range Radar Altimeter

LR-D1

User Manual

D00.01.13
10/29/2025

Revision History

Version Number	Date	Authors	Notes
D00.00.01	Sep 13, 2018	Hao Liu, Liqiang Ren, Zhenyu Hu	Initial Draft
D00.00.03	Oct 17, 2018	Zhenyu Hu	Add section 7 - Known Issues
D00.01.01	Nov 5, 2018	Andrew Megaris	Technical Revision
D00.01.02	Dec 6, 2018	Zhenyu Hu	Spec updates
D00.01.03	Jan 14, 2019	Zhenyu Hu	Data protocol update
D00.01.04	May 17, 2021	Zhenyu Hu	From the original version of V2.2
P00.01.04	June 2, 2021	Ethan Perrins	Public Version
D00.01.05	June 18, 2021	Zhenyu Hu	From the original version of V2.4
D00.01.06	September 17, 2021	Camron Myers	Added 8. Firmware Change Log, 9. Firmware Update Instructions, and 10. Test Tool GUI.
D00.01.07	Oct 7, 2021	Zhenyu Hu	From original version of V3.0 for firmware 3.223.7/3.0.7
D00.01.08	Feb 20, 2023	Camron Myers	V2.2 Hardware Changes
D00.01.09	July 12, 2023	Zhenyu Hu Camron Myers	Alert Information Changes; HW2.2 Changes
D00.01.10	Aug 16, 2024	Zhenyu Hu	Update info by the

			released hardware and software in August 2024
D00.01.11	Oct 11, 2024	Pedro Lopez, Zhenyu Hu	Update info by only supporting released firmware for hardware V2.2
D00.01.12	July 10, 2025	Nash Sloan, Pedro Lopez, Zhenyu Hu	Update info by Adding Out of Range behavior indication Byte
D00.01.13	October 29, 2025	Zhenyu Hu	Firmware updated to v19.0.0.1

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1. Product Description

1.1 LR-D1

The LR-D1 Radar Altimeter uses the principles of radio detection and ranging to determine the altitude of the aircraft. A microwave signal is transmitted out of the sensor, reflects off the target, and is received by the sensor. Distance and horizontal velocity from the sensor to the terrain (altitude) are derived by the difference in time from when the signal is sent from the sensor to when the signal is received by the sensor.
The LR-D1 is enclosed in a black metallic case, the radome, which is conducive to radar telemetry.

1.2 Compliances

FCC and CE certifications pending.

2. Installation Guidelines

2.1. Mounting Angle

When mounting the device, the radio wave emitted from the LR-D1 must be **perpendicular** to the ground below the aircraft. There should not be any angle of inclination in any direction while the device is fastened to the aircraft.

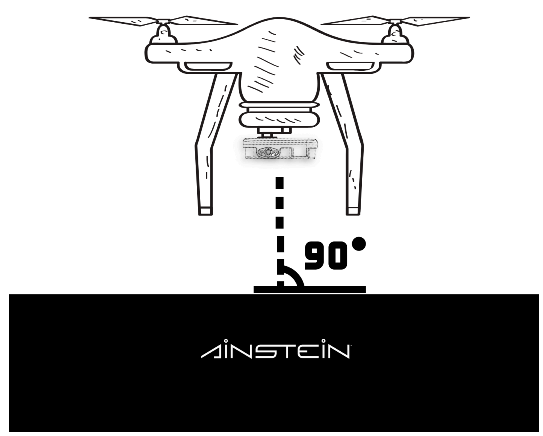

Figure 1

2.2 Mounting to an Aircraft

The device should be secured to the aircraft, where it is not free to move in any direction.

2.3. Line of Sight Clearance

Keep the face of the radar clean, and do not cover it with any additional materials. Any coatings, coverings, and modifications to the radome can degrade the performance of the radar device.
Additionally, keep any unexpected objects out of the radar's FoV (Field of View). Obstructions to the LR-D1's field of view will cause a decrease in the performance of the radar. It is highly recommended that the LR-D1 be mounted on the underside of the aircraft far away from the landing gear, other aircraft structures, or other equipment.

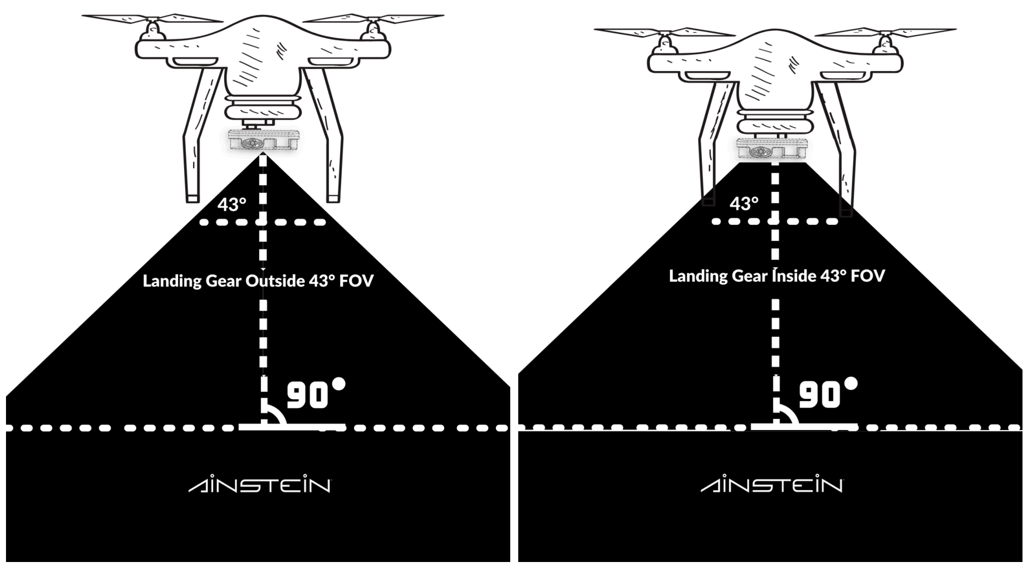

Figure 2

2.4. Integration Requirements

The LR-D1 outputs altitude measurements and signal-to-noise ratio (SNR) measurements when operational. When integrating the LR-D1 radar altimeter, it is necessary to use SNR measurements and Out of Range Behavior Indication in conjunction with malfunction alerts (see appendix 1) to properly filter out erroneous altitude values.

A filtering algorithm should be used to estimate vehicle position, velocity, and angular orientation based on rate gyroscopes, accelerometer, compass, GPS, airspeed, and barometric pressure measurements in addition to the recorded LR-D1 measurements. Sensor redundancy is heavily advised for the LR-D1.

3. Operational Requirements

The LR-D1 will perform optimally if the operational requirements below are fully satisfied. Failure to meet the operational requirements may cause a decrease in performance, accuracy, or reliability of the LR-D1 altimeter and is not advised.

3.1 Obstruction to the LR-D1 Field of View

Objects or aircraft structures that are located within the LR-D1's conical field of view which obstruct the radar's view of the ground may cause multipath reflections or other degradative phenomena to occur. The LR-D1 should be mounted a safe distance away from the landing gear and any other components of the aircraft below it.

3.2 Excessive Pitch/Roll

Pitch and roll angles that exceed 21.5° and 15° respectively may also cause the performance of the LR-D1 to worsen. This sensitivity increases with altitude.

3.3 Terrain

Terrain with poor reflectivity may cause the performance of the LR-D1 to worsen. Flying at the limit of the LR-D1's range over dry, loose soil, such as tilled farmland, or sand is not recommended.

Terrain with high reflectivity may cause the LR-D1 to output abnormally high values at low altitudes due to high saturation.

3.4 Power Source

Any power source used to operate the LR-D1 that does not provide the minimum power and voltage can worsen the performance of the device or cause its operation to stop altogether.

3.5 Orientation of LR-D1

The LR-D1 must be mounted upon the aircraft in such a manner as to be fully horizontal, its radome directly facing the ground. Any angle of inclination may degrade performance.

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3.6 Minimum and Maximum Operating Altitude

Operating the LR-D1 at altitudes below 1.4 meters and above 500 meters will result in a degradation of performance and potentially erroneous measurements. Also, any reading at an altitude of greater than 655.35m would be considered an error or overflowed reading (see appendix 1). **When the actual altitude is too high or too low, for example, above concrete, the radar can operate normally up to 800m or even higher. It can also operate normally below than 1.4m, for example, when the mounting height on some aircrafts are on the surface or at low heights. The LR-D1 will report data12 (Out of Range Indication, see Table 2 in the Section 5 for details) is invalid to indicate there is no valid target detected.**

4. Technical Data:

Table 1: Specification

Frequency Band	24 GHz
Bandwidth	250 MHz
Power Consumption	< 11.00 W
Operating Voltage	10 - 30 V
Altitude Range	1.4m~500m (1)
Altitude Precision	±0.3644m (2)
Update Rate	40Hz
Detection Angle Range	Azimuth 43°, Elevation 30° (3)
Maximum Velocity	± 30 m/s in elevation
Temp. Range	-40℃~ +60℃
Dimensions	<112mm*102.5mm*31mm (mounting bracket is NOT included)
Weight	315 g (excluding external connector cable)
IP Rating	Built to the requirements of IP67 (Test pending) (4)
Vibration Rating	IEC 60068-2-6.1995 sine vibration 5g XYZ three axis
Shock Rating	IDT IEC 68-2-27:1987 half sine shock 20g XYZ three axis

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ESD Rating	IEC 61000-4-2.2008 8K/15K contact/air B class

Note:

1. Radar data may vary over different terrains when the radar is out of its detection range. Usually, we recommend mounting the LR-D1 at a minimum height of at least 1.4m.

2. Range detection might be limited by terrains, pitch, and roll of aircraft, etc. The range precision here only indicates the lab experiment/calibration result in the ideal case.

   LR-D1 altitude data report step size could be smaller than this precision because of the post-processing.

3. Based on mm-wave radar specs, a large angle of pitch and roll would bring error for detection. Under the same measurement circumstance, larger angles by aircraft bring more error.
4. IP rate here only focuses on the radar itself. This rating does not cover any cabling interface.

5. UART Data Protocol For LR-D1:

- Protocol: UART
- I/O Standard: RS-232 (Default) and RS-422 (Per Request)
- Baud Rate: 115200 b/s
- Data length: 8 bits, plus one start bit and one stop bit, and no parity bit

Caution! If the data received for the malfunction alert code is not equal to 0x0000 (Normal function) for data 5 and data 6 in the table below, then please check the malfunction alert code table in the Appendix 1 for details. .

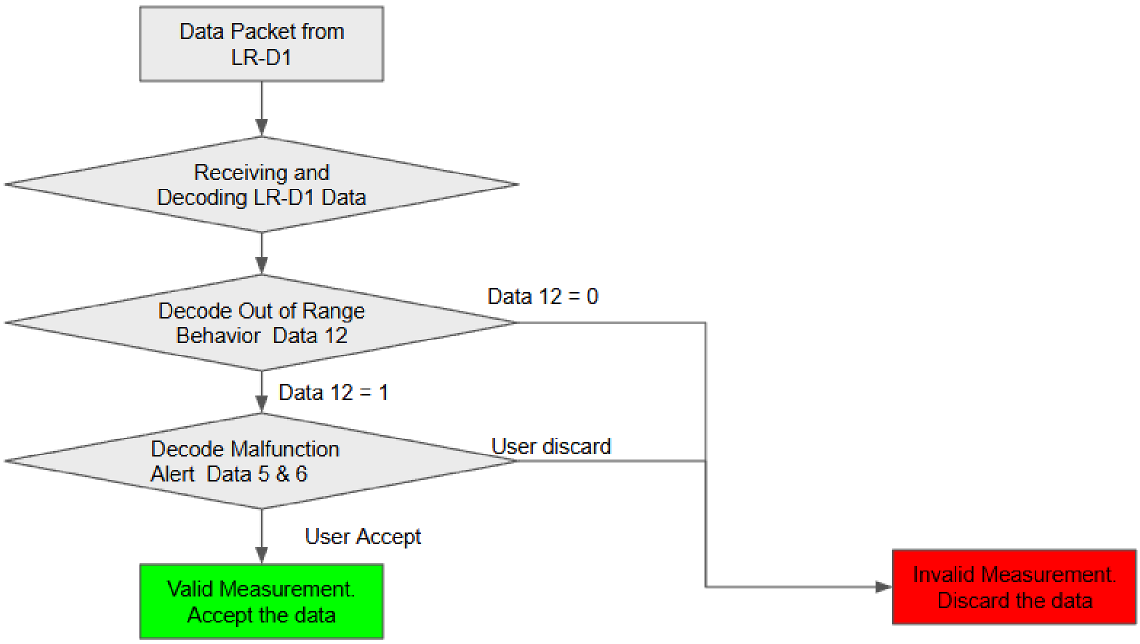
Table 2: Data Packet Definition

From	LR-D1	To	Receiver
Byte	Data	Note	
data1	0xEB	Packet Header MSB (Most Significant Bits)	
data2	0x90	Packet Header LSB (Least Significant Bits)	
data3	deviceID	Device ID Byte (0x00)	
data4	0x1C	Data packet length; Fixed as 28 Bytes	
data5	0x00: Normal Others: Malfunction[1]	Malfunction Alert MSB	
data6	0x00: Normal Others: Malfunction	Malfunction Alert LSB	
data7	high1_h[2][6]	Object 1 Altitude MSB	
data8	high1_l[2][6]	Object 1 Altitude LSB	
data9	snr1	Object 1 SNR	
data10	speed1_h[3]	Object 1 Velocity MSB	
data11	speed1_l[3]	Object 1 Velocity LSB	
data12	Out of Range Indication (Altitude Health) [4]	0: Invalid Altitude 1: Valid Altitude	
data13	Reserved [5]	0xFF	
data14		0xFF	
data15		0xFF	
data16		0xFF	
data17		0xFF	
data18		0xFF	
data19		0xFF	
data20		0xFF	
data21		0xFF	
data22		0xFF	
data23		0xFF	
data24		0xFF	

data25		0xFF
data26		0xFF
data27		0xFF
data28		0xFF
data29		0xFF
data30		0xFF
data31		0xFF
data32	checksum	Checksum: (data4+data5+…+data29+data31) bitwise-AND with 0xFF

Note:

1. Please see Appendix 1 for details about malfunction information.
2. Altitude Data Parse: Altitude = (high_h * 256) + high_l; unit: **0.01 m (cm)**; Type: **Unsigned**
   <u>**Note:**</u> If 'Out of Range Indication' is 0, the altitude will show '0.'
3. Velocity Data Parse: Velocity = (speed _h * 256) + speed _l; unit: **0.01m/s**; Type: **Signed**
   SNR Data Parse: SNR = snr1; unit: dB; Type: Unsigned
   <u>**Note:**</u> If 'Out of Range Indication' is 0, the velocity and SNR will show '0.'
4. Out of Range Behavior (Radar Measurement Health):
   - 0: Invalid Altitude, the radar reported altitude health is 'Invalid'.
   - 1: Valid Altitude, the radar reported altitude health is 'Valid'.
5. Reserved Byte: Not applied in current version and default as 0xFF.
   <u>**Note:**</u> In the Beta version firmware, Reserved Bytes may be filled with values other than 0xFF for evaluation purposes.
6. We use a 16-bit integer to indicate the LR-D1's altitude reading in centimeters, therefore the maximum valid altitude would be 65535 cm (655.35 m). If the actual altitude is greater than that, altitude reading overflow would occur, and will report 'Altitude reading overflow' Malfunction Alert.
7. There are two data sections in the LR-D1 that are related to the Radar Measurement's validity (See the example in the flow chart):
   a. Data 12 Altitude Health: This is an absolute criteria. Users MUST NOT use radar's measurement if this data reports 'Invalid'
   b. Data 5~6 Malfunction: This is a relative criteria. Users SHOULD consider the application to determine whether to keep using radar's measurement or discard it. For example:
      i. 'Altitude reading overflow' - Users might discard the altitude reading or consider the altitude reading plus 65535 cm.
      ii. 'Voltage alert' - Users might discard the radar's measurement reading or keep using it temporarily and contact Ainstein for more details.


Data Packet from LR-D1
Receiving and Decoding LR-D1 Data
Decode Out of Range Behavior Data 12
Data 12 = 0
Data 12 = 1
Decode Malfunction Alert Data 5 & 6
User discard
User Accept
Valid Measurement. Accept the data
Invalid Measurement. Discard the data

6. Mechanical Drawing:

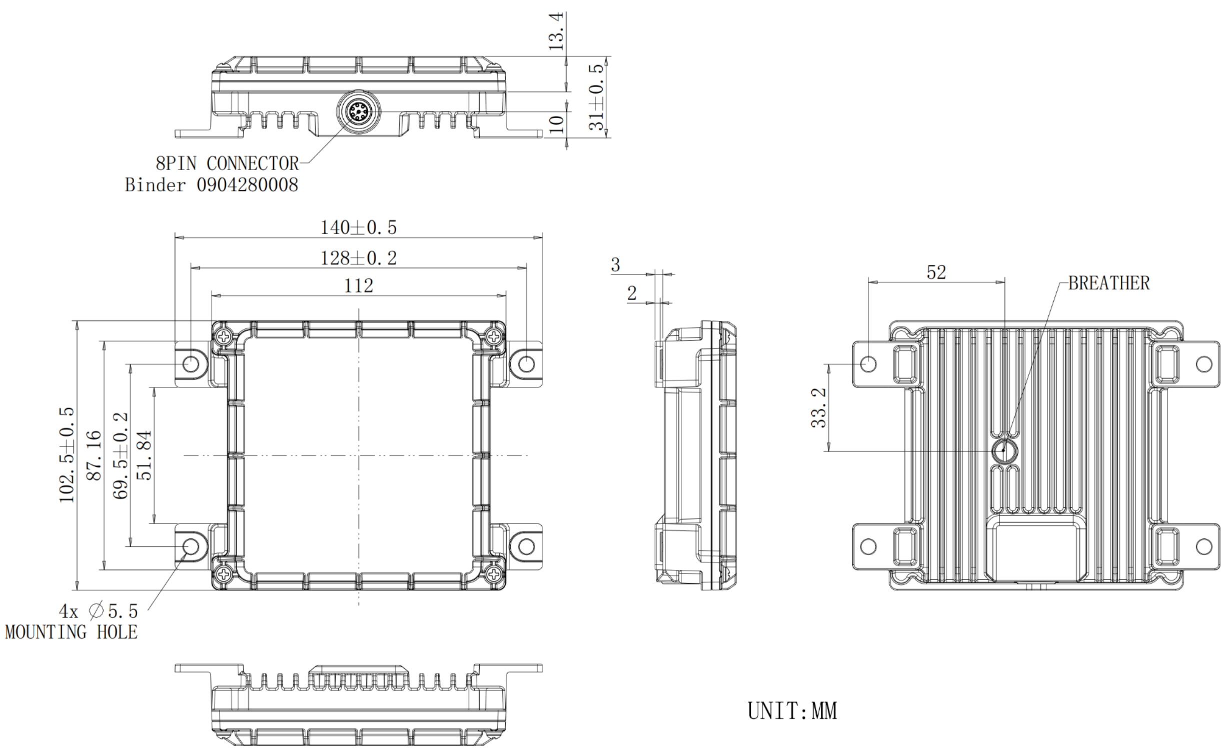


Figure 3: Dimensions of LR-D1 (Units: mm)

7. Radiated Emissions

Maximum Transmit Power (EIRP)	32 dBm

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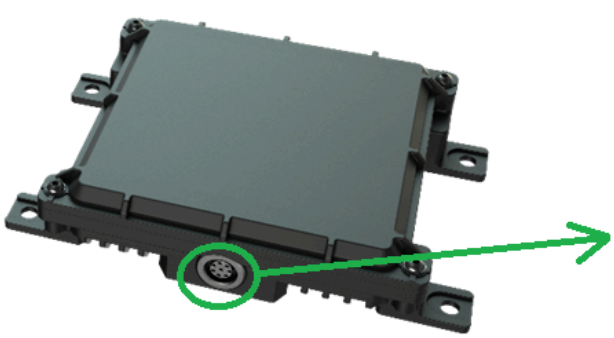
8. Hardware Interface:

Table 3: Pin Out Definition

Pin	Wire Color	Pin Name	Function	Note
1	Red	VCC	Input Voltage	10 - 30 V Power < 11W
2	Red	VCC	Input Voltage	10 - 30 V Power < 11W
3	Blue	RS-422 T+ / RS-232 T	RS422: TX+ RS232: TX	Default Setting: RS232
4	Brown	RS-422 T- / RS-232 R	RS422: TX- RS232: RX	Default Setting: RS232
5	White	RS-422 R+	RS422: RX +	Leave unwired for RS232
6	Green	RS-422 R-	RS422: RX-	Leave unwired for RS232
7	Black	GND	Ground	
8	Black	GND	Ground	

Note:
LR-D1's default hardware interface is RS-232. If the RS-422 interface is required, please contact Ainstein for assistance.



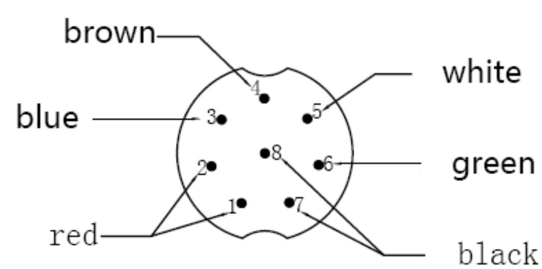


Figure 4: LR-D1 Pinout Diagram

9. Cabling Diagram:

The default mating connector used for the LR-D1 is the BINDER 99-0425-10-08 circular connector, pictured below in Figure 4.

Figure 4: 6-Wire Cable with BINDER 99-0425-10-08 Connector

The BINDER 99-0425-75-08 90° connector can be provided for an additional cost at the time of purchase. See Figure 5.

Figure 5: 6-Wire Cable ith BINDER 99-0425-75-08 Connector

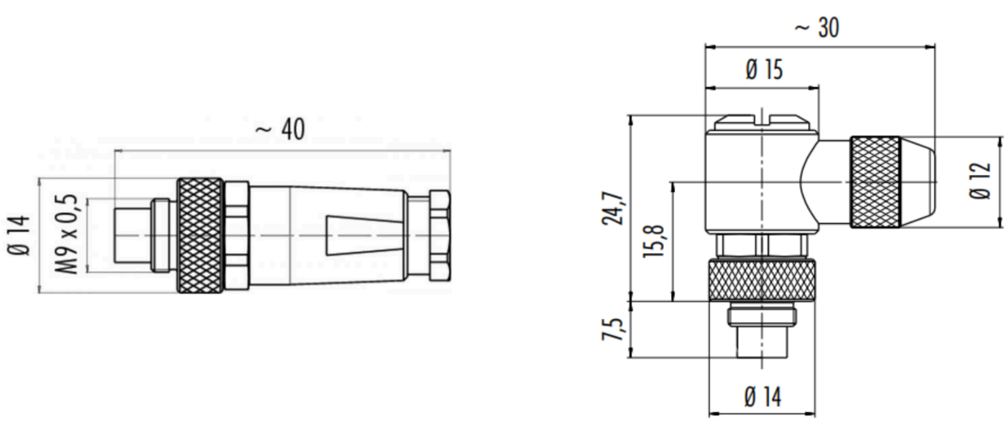


Figure 6: BINDER 99-0425-10-08 & BINDER 99-0425-75-08 Connector Dimensions.

10. Known Issues

LR-D1 is a product still in development. Table 4 lists the known issues that will be addressed in later revisions.

Table 4: LR-D1 Known Issues

Issue ID	Description	Notes
1	If the LR-D1 is mounted at the position of < 1.4m, it would report 'Unhealthy' in the Out of Range Indication Byte or 1.4 m of range reading	• Ignore all radar detection readings if Out of Range Indication reports Unhealthy • DO NOT consider altitude data while on the ground in this case • If the altitude is too close, it means the radar reported altitude health is close to the surface or target. Please consider it as an **<u>'Invalid'</u>** report.
2	Altitude data from radar may have unexpected or incorrect readings if the application scenario is indoors. The reason is the multipath reflection of radar is complicated indoors, it may give unexpected or incorrect readings under this circumstance.	• DO NOT apply radar in any indoor application scenario
3	Altitude data from radar may have unexpected or incorrect readings if aircraft pitch and/or roll are beyond the radar's detection angle (Azimuth 43°, Elevation 30°). The ideal case is keeping the radar perpendicular to the ground.	• Add a gimbal with radar mounting • DO NOT consider any altitude data as valid if the aircraft's pitch and/or roll are beyond the radar's detection angle.
4	Voltage alert is monitoring multiple node voltage on the radar's hardware, some of their	• Check the input voltage

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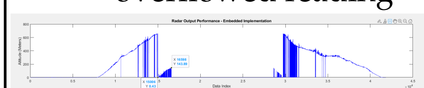
	thresholds might be set too stringent and trigger alert frequently in some scenario, we are evaluating to extend the acceptable thresholds to resolve it	• Cool down the device and improve the heat dissipation • Notify Ainstein if this alert still reports frequently after trying out above solutions

5	Overflowed Reading	• The altitude for the two spots in the below figure are:
		  ○ 655.35 + 0.43 = **<u>655.78 m</u>**
		  ○ 655.35 + 143.89 = **<u>799.24</u>** m
		• It is user's decision to discard or accept the overflowed reading
		

11. Firmware Change Log:

The LR-D1 current firmware version is 19.0.0.1. Its development is noted in the following table. For older FW versions, reach out to Ainstein.

Table 5: LR-D1 Firmware Change Log

Firmware Version	Release Date	Notes
18.0.1.1 [Obsolete]	01/29/2023	• First released firmware on LR-D1 hardware V2.2
18.0.1.2 [Obsolete]	02/02/2023	• Maintain performance from 18.0.1.1 • Add production mode for manufacturing
18.0.1.3	05/11/2023	• Maintain performance from 18.0.1.2 • Fixes the voltage monitor reading consistency issue
19.0.0.0 [Obsolete]	6/30/2025	• Add Out of Range Indication in Data 12
19.0.0.1	10/29/2025	• Fixed a case of serial port stop reporting data in v19.0.0.0

12. Firmware Update Instructions:

Remote Firmware Update

To update the firmware:

1. Open the GUI application
2. Select the Serial Port connected to the LR-D1.
3. Set the Baud Rate to 115200.
4. Select the version to “Gen2”, it indicates LR-D1 V2.2
5. Click Browse and select the desired .bin file.
   <u>Note:</u> From firmware v18.x.x.x, both RS-232 and RS-422 units use the same firmware bin file.
6. Click Load to update the firmware on the LR-D1.
7. Upon successful completion of the firmware update, the device will automatically restart.

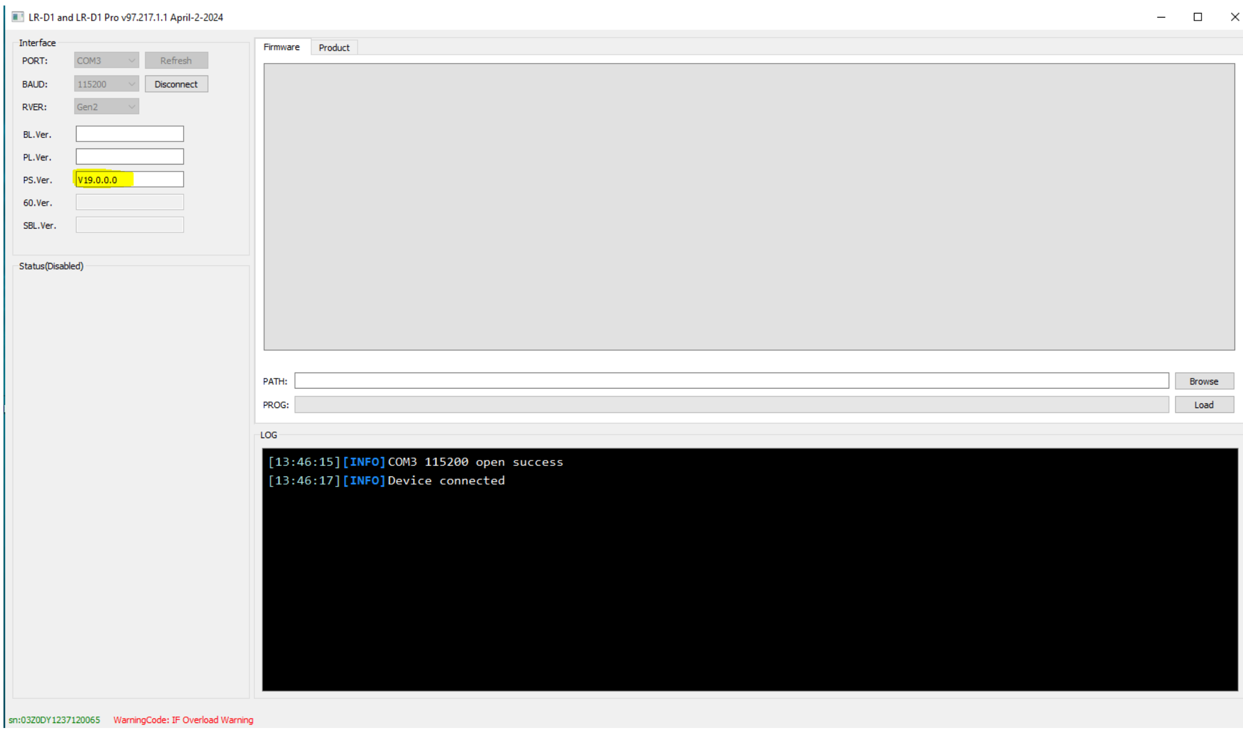


Figure 7: LR-D1 “main” Firmware Update GUI

13. Contact US

ADDRESS:
1421 Research Park Dr.,
Lawrence, KS 66049 USA

EMAIL:
hi@ainstein.ai

PHONE:
785-424-7194

Appendix 1: Malfunction Alert Information List

Caution! If the data received for the malfunction alert code is not equal to 0x00 (Normal function) for data 5 and data 6 found in **Table 2: Data Packet Definition**, then please do not use the data packet and filter it out.

There are four types of alerts open to the end users, the type of alert can be distinguished by its special code below.

There are two Bytes of data reserved for the Malfunction Alert. All four types of alerts are represented by a bit in this Byte. For example:

- 0x0001 (0b0000000000000001) of the temperature alert is represented by the bit-0 of the Malfunction Alert Byte;
- 0x0002 (0b0000000000000010) of the voltage alert is represented by the bit-1 of the Malfunction Alert Byte;
- 0x4000 (0b0100000000000000) of the IF signal saturation alert is represented by the bit-6 of the Malfunction Alert Byte;
- 0x8000 (0b1000000000000000) of the altitude reading overflow alert is represented by the bit-7 of the Malfunction Alert Byte;

Also, some of these Malfunction Alerts might be combined. For example:

- 0x0003 (0b0000000000000011) of the temperature alert and the voltage alert
- 0x4003 (0b0100000000000011) of the temperature alert, the voltage alert, and the IF signal saturation alert

For obsolete Firmware (before v18.x.x.x), please contact Ainstein for a solution.

For Firmware from v18.X.X.X:

Malfunction Alert Code	Malfunction Alert Info	Possible Reasons	Suggestion
0x0001	MCU Temperature alert	Error in the device or ambient temperature is out of LR-D1's operational temperature	Stop using and contact Ainstein

0x0002	MCU Voltage alert	Error in device	Stop using and contact Ainstein for evaluation.
0x0010	IF Temperature alert	Error in the device or ambient temperature is out of LR-D1's operational temperature	Improve the heat dissipation. If the issue still remain, stop using and contact Ainstein Note: Long time operation in the static scenario (e.g. indoor testing) might bump the temperature significantly and report this alert code
0x0080	IF signal saturation alert	The reflection signal received by LR-D1 is overloaded	Occurs when LR-D1 is within 1.4m of the ground. Check if there is any other target within LR-D1's FoV
0x0100	Software alert	Malfunctions occur in the software	Stop using and contact Ainstein
0x0400	Altitude reading overflow alert	The altitude reading is greater than the maximum integer of 16-bit (655.35 meters)	Occurs when LR-D1 detects the altitude is greater than 655.35 meters.
0x8000	Voltage alert	Error in device by over-voltage or under-voltage	Check the supply-voltage. Recommended voltage range is 12~28V Note: Minimum of 10V of supply-voltage is sufficient to operate the device, however it might trigger this alert code.